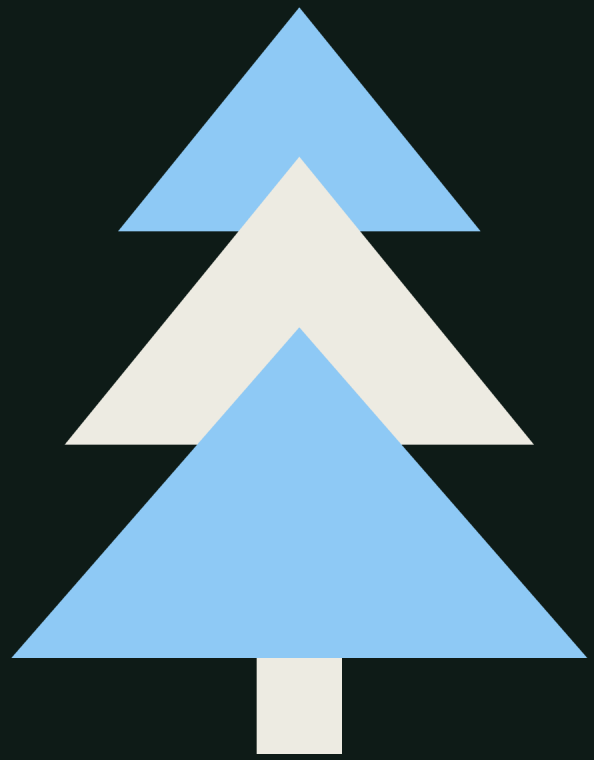
PARENTHESIS PROGRAMMING LANGUAGE

ForestPPL

A lightweight scripting interpreter in C++ with a plug-in library model

MIT license · Windows, Linux port in progress



What is ForestPPL?

Console application

Written in C++ for Windows, with a Linux port being verified. Interactive prompt or script files.

Powered by PPL.DLL

Runs the Parenthesis Programming Language: a small language where every call is a pair of parentheses.

Extensible

Libraries are plug-in DLLs loaded on demand with import X;

Two notations, one language

.scr · infix

Familiar, readable

```
write#(1+3);
```

.ppl · prefix

Native, fully parenthesized

```
write(+(1)(3))
```

Convert with createpplcode(file.scr)(file.ppl)

SUPPORTED DATA TYPES

var	const	array	storage	node	structure

How it works

Input

Prompt or script file

Command queue

Producer–consumer

Processing thread

Parser and AST traversal

Libraries

Plug-in DLLs

A ScopeManager tracks variables and scopes while the processing thread runs commands one by one.

Configuration | **Common and User Functions** | **Numerous samples**

Running scripts

`>`	Interactive prompt: type commands directly
`rc examples\scr\sieve3.scr`	Run a script inside the current session
`ForestPPL script.scr`	Run from the command line
`checksyntax yes`	Validate syntax before execution

stay_interactive yes|no controls whether the prompt stays open after a script finishes.

Built-in libraries

Directory
Folders and paths

Math
Numeric functions

String
Text processing

File
Read and write files

Console
Colors and output

StorageLib
Multi-dimensional arrays

Random
Random numbers

NumPPL
Numerical arrays

Requests
Web requests

Load any of them with import String;

A small script

```
array arr[] = {1,2,3,4,5};
array.remove(arr,1);
writearray arr row;
```

OUTPUT

```
1,3,4,5
```

Remove an element by index, then print the array as a row.

Storage: multi-dimensional arrays

Jagged arrays of any depth, each dimension resizable on demand.

Functions are called with the storage. prefix: Push, Sum, sresize, sdimsize, swrite and more.

```
import storage;
storage s;
s.Push(10);
s.Push(20);
write(s.Sum());  // 30
```

NumPPL: NumPy-style arrays

26 functions for numerical work on Storage arrays.

Reshape, Transpose, Dot, Det, Inv, Solve and more.

```
a = Reshape(1..6)(2)(3);
b = Transpose(a);
c = Dot(a)(b);
write(Det(c));
```

```
14 32 ; 32 77   det = 54
```

Write your own library

1

Describe

Declare functions in a .cfg file; CodeGen produces the glue code.

2

Implement

Write each FuncX in C++ and register it with AddToKeywordDictionary.

3

Import

Export Name_CreateInstance from the DLL, then import Name;

Tooling

Syntax checking

checksyntax yes catches errors before a script runs.

VS Code extension

Editing support for .scr and .ppl files.

Documentation

HTML reference and a PDF tutorial.

Debugging

debugppl, traceppl, suspend/resume, start/stop and log show what the interpreter is doing.

Open to contributors

License · MIT

Platforms · Windows, Linux

Tutorial

ForestPPL SFX application

Questions and ideas · GitHub Discussions

`github.com/okogosov/ForestPPL`